Year 10 Science
Physical Sciences
Worksheet - Newton's Laws

NEWTON'S FIRST LAW OF MOTION

"An object continues in its state of rest or uniform motion in a straight line unless acted on by a net external force".

This law is also known as the *law of inertia. Inertia is the tendency of a body to remain in its state of motion.*

The greater the mass of a body the greater is its inertia.

FORCE

A force may be defined as that which produces or tends to produce a change in the state of rest or motion of a body.

NEWTON'S SECOND LAW OF MOTION

"When a net force acts on a body the body is accelerated in the direction of the force. The acceleration is directly proportional to the force and inversely proportional to the mass of the body".

Mathematically this law can be expressed as:

$$a \propto \frac{F}{m}$$

or $F = k\,m\,a$ where
F = net force
k = proportionality constant
m = mass of the body
a = acceleration produced

When F, m and a are expressed in SI units $k = 1$ and the law may be written in its common form:

$$\boxed{F = m\,a}$$

UNITS OF FORCE

The unit of force is the *newton (N)* and is defined such that *a net force of 1 N acting on a mass of 1 kg produces an acceleration of 1 m s^{-2}.*

$$\begin{aligned} F &= m\,a \\ &= (1\ \text{kg})(1\ \text{m s}^{-2}) \\ \therefore 1\ \text{N} &= 1\ \text{kg m s}^{-2} \end{aligned}$$

TYPE EXAMPLE 21: What force acting on a mass of 8 kg will produce an acceleration of 1.5 m s^{-2}?

$F = ?$
$m = 8$ kg
$a = 1.5$ m s^{-2}

$$\begin{aligned} F &= ma \\ &= 8 \times 1.5 \\ &= 12\ \text{N} \end{aligned}$$

i.e. a force of 12 N is required (in the direction of the acceleration).

TYPE EXAMPLE 22: A force of 50 N is applied to a mass of 12.5 kg. What is the acceleration produced?

$$F = 50\text{ N} \qquad F = ma$$
$$m = 12.5\text{ kg} \qquad \therefore a = \frac{F}{m}$$
$$a = ? \qquad = \frac{50}{12.5}$$
$$= 4\text{ m s}^{-2}$$

i.e. an acceleration of 4 m s^{-2} is produced (in the direction of the force).

SET 15: NEWTON'S SECOND LAW OF MOTION I

1. A mass of 6 kg is accelerated at 3.5 m s^{-2}. What net force is applied?
2. Determine the magnitude of the force which when acting on a mass of 2.5 kg produces an acceleration of 5 m s^{-2}.
3. What force must act on a car of mass 2 000 kg to produce an acceleration of 1.6 m s^{-2}?
4. Find the acceleration produced when a force of 200 N acts on a body of mass 40 kg.
5. A force of 36 N acts on a 24 kg mass. What acceleration is produced?
6. What acceleration is produced when a net force of 16 N acts on a mass of 48 kg?
7. Determine the acceleration resulting when a force of 625 N acts on a body of mass 125 kg?
8. A force of 72 N produces in a body an acceleration of 1.2 m s^{-2}. What is the mass of the body?
9. What is the mass of a car which is accelerated at 2 m s^{-2} when acted on by a force of 2 800 N?
10. A force of 3 N is applied to a block on a smooth horizontal table and produces an acceleration of 11 m s^{-2}. What is the mass of the block?

SET 15

1. 21 N **2.** 12.5 N **3.** 3 200 N **4.** 5 m s^{-2} **5.** 1.5 m s^{-2}
6. 0.33 m s^{-2} **7.** 5 m s^{-2} **8.** 60 kg **9.** 1 400 kg **10.** 0.27 kg

WEIGHT AND MASS

Mass refers to the quantity of matter in a body and is measured in kilograms (kg).

Weight is the earth's gravitational force of attraction on a body and is directed towards the centre of the earth. As weight is a force it is measured in *newtons (N)*.

Freely falling bodies are accelerated at 10 m s^{-2} (g) near the earth's surface.

A mass of 1 kg falling freely near the surface of the earth will be accelerated at 10 m s^{-2}. The gravitational force of attraction producing this acceleration can be calculated by:

$$
\begin{aligned}
F &= ma \\
&= (1 \text{ kg})(10 \text{ m s}^{-2}) \\
&= 10 \text{ kg m s}^{-2} \\
&= 10 \text{ N}
\end{aligned}
$$

From this it can be seen that the gravitational force of attraction on a 1 kg mass is 10 N and from the definition of weight, the weight of a mass of 1 kg is 10 N downward.

As all bodies are accelerated at g near the earth's surface the weight of a body can be determined from:

$$F = ma$$

$$\boxed{F_w = mg}$$

where: F_w = weight (N)
m = mass (kg)
g = acceleration due to gravity

TYPE EXAMPLE 25: Determine the weight of a 6 kg mass.

$m = 6$ kg
$g = 10$ m s^{-2}
$F_w = ?$

$$
\begin{aligned}
F_w &= mg \\
&= 6 \times 10 \\
&= 60 \text{ N}
\end{aligned}
$$

i.e. the weight of a 6 kg mass is 60 N downward.

SET 17: WEIGHT

Use g = 10 m s^{-2} for acceleration due to gravity near the earth's surface.

1. What is the weight of a:
   (a) 52 kg mass? (b) 2.6 kg mass?
2. Determine the mass of bodies with the weights:
   (a) 350 N (b) 5 N
3. Find the weight of a student of mass 48 kg.
4. What is the weight of a 1 000 kg motor car?
5. What is the mass of a crate which weighs 1 200 N?
6. A vehicle of mass 800 kg has a horizontal force equal to its weight applied to it. Find the:
   (a) weight of the vehicle
   (b) acceleration produced.
7. A body of mass 4 kg at rest is acted upon by a force equal to half its weight. Determine after 3 s, the:
   (a) velocity
   (b) displacement.
8. A body is suspended on a spring balance and shows a weight of 20 N. This same balance and body on the surface of the moon would show 3.2 N. What is the acceleration due to gravity on the surface of the moon?

SET 17

1. (a) 520 N (b) 26 N **2.** (a) 35 kg (b) 0.5 kg **3.** 480 N **4.** 10 000 N
5. 120 kg **6.** (a) 8 000 N (b) 10 m s^{-2} **7.** (a) 15 m s^{-1} (b) 22.5 m
8. 1.6 m s^{-2}